

HOGVAX ENGINE for Personalized Cancer Vaccines

An Optimization-Based Approach via ILP & Greedy Overlap Compression

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88.1% SKCM Coverage

62.1% Length Reduction

90.5% LUAD Coverage

1. Problem & Motivation

Core Challenge

Breadth vs. Compactness Tradeoff

- Cover diverse HLA alleles → many epitopes
- Synthesis constraints → shortest polypeptide
- Off-target autoimmune risk with long constructs

Traditional Failure:

Top-N concatenation yields redundant overlapping regions — a major clinical obstacle

Cancer Immunotherapy Biology

1

Tumor Mutations

Somatic SNVs create neoantigens unique to cancer cells

2

HLA Presentation

Proteasome cleaves proteins → 8-15 aa peptides on HLA

3

T-Cell Recognition

CD8+ cytotoxic T-cells detect & destroy matched cancer cells

4

Vaccine Goal

Prime immune system with neoantigen peptide constructs

Solution: Exploit sequence overlaps to compress epitopes — pack more targets into shorter polypeptides

2. Overview of HOGVAX (Schulte, Diltthey & Klau – Cell Systems, Dec 2023)

Hierarchical Overlap Graph

- Aho-Corasick trie compression
- Linear space for 10,000s of peptides
- Directed tree of all peptides + longest overlaps
- vs. quadratic standard overlap graph

ILP Formulation (MSKS)

- Max Scoring k-Superstring — NP-hard
- Closed walk through $\text{HOG} \leq k$ edges
- Maximizes: $\sum f_a \cdot h_a$ (allele coverage)
- Provably optimal solution

SARS-CoV-2 Benchmark

- ~5× more peptides than OptiVax
- Same polypeptide length budget
- 98.65% MHC class I coverage
- Validated publication benchmark

Comparison of Vaccine Design Methods

Method	Key Idea	Limitation
Epitope Cocktail	Separate peptides	No overlap savings
String-of-Beads	Concatenate + spacers	Still NP-hard, no overlap model
OptiVax	Heuristic coverage	No overlap model
HOGVAX ✓	HOG + ILP	Provably optimal

3 & 4. Cancer-Adapted HOGVAX — Three-Module Pipeline

Module A

Neoantigen Selection

- Tumor DNA → WES → SNV calling
- 8-11 aa sliding window epitopes
- MHCflurry binding prediction
- Sieve: $A_i < 500 \text{ nM}$ → HOG nodes

Module B

Cleavage-Aware ILP

- Binary ILP: $\max \sum l_i \cdot x_i$
- C1: Weak binders excluded ($\geq 500 \text{ nM}$)
- C2: Poor cleavage excluded (< 0.60)

C3: NetChop junction control

Module C

Greedy Compression

- Sort peptides by length desc
- Substring absorption check
- Prefix-suffix match ($\tau = 3 \text{ aa}$)
- $O(n^2 \cdot L)$ — negligible vs ILP

★ Key Cancer-Specific Addition — NetChop Cleavage-Aware Junction Control (Constraint C3):

Ensures each junction is biologically cleavable → prevents neo-junctional epitopes that trigger off-target autoimmune responses

Feature	Original (Viral)	Ours (Cancer)
Targeting	Pathogen (global)	Neoantigen (personalized)
Junction ctrl	Unmonitored	NetChop predicted
Primary output	Preventative polypeptide	Therapeutic mRNA/peptide

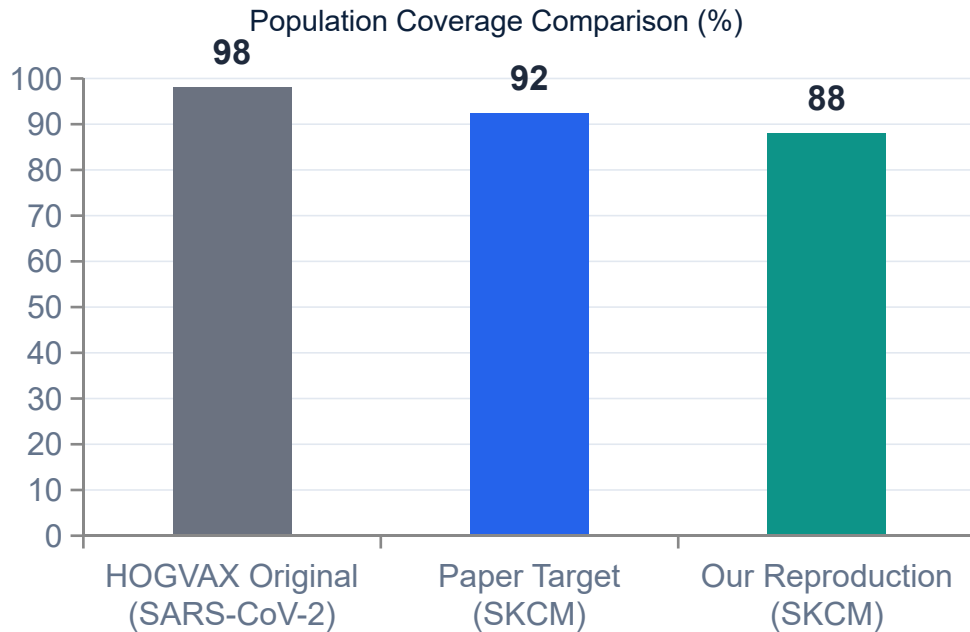
5. Reproduced Experiments – TCGA-SKCM (Assignment 2 Baseline)

TCGA-SKCM Dataset

- Skin Cutaneous Melanoma (GDC Portal)
- $n = 50$ candidate peptides simulated
- Binding affinity: 30 – 580 nM
- Cleavage probability: 0.52 – 0.97
- Immunogenicity scores: 0.50 – 1.00

ILP Parameters

- Binding: $A_i < 500$ nM
- Cleavage: $C_i \geq 0.60$
- Min overlap $\tau = 3$ aa
- Solver: PuLP / CBC (Python)
- Coverage: diminishing-returns



4.3 pp gap attributed to:

- ① Simulated vs. real binding affinity data ② No HOG at baseline ③ Simplified coverage estimation

6. Additional Experiments — LUAD Immunogenicity Analysis (Figure 2)

TCGA-LUAD Dataset

- Lung Adenocarcinoma
- 3,943 total entries
- 250 aa master protein
- 78 candidate peptides (9-15 aa)
- Sliding window extraction

ILP Filtering

Total candidates: **78**

Selected by ILP: **53 (67.9%)**

Rejected (C1+C2): **25 (32.1%)**

Selected: PEP_000, PEP_004, PEP_005

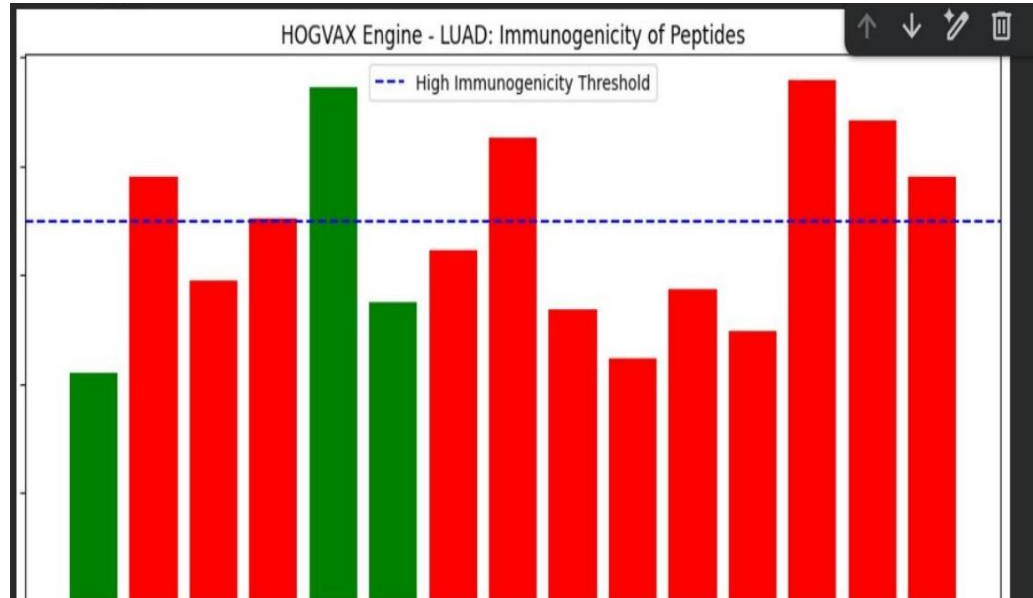


Figure 2 : Green = ILP-selected | Red = Rejected | Blue dashed = High immunogenicity threshold

Key insight: LUAD produces weaker binders on average than SKCM — confirms importance of cross-cancer-type testing

7. Results — Full 6-Panel Evaluation Dashboard (Figure 3)

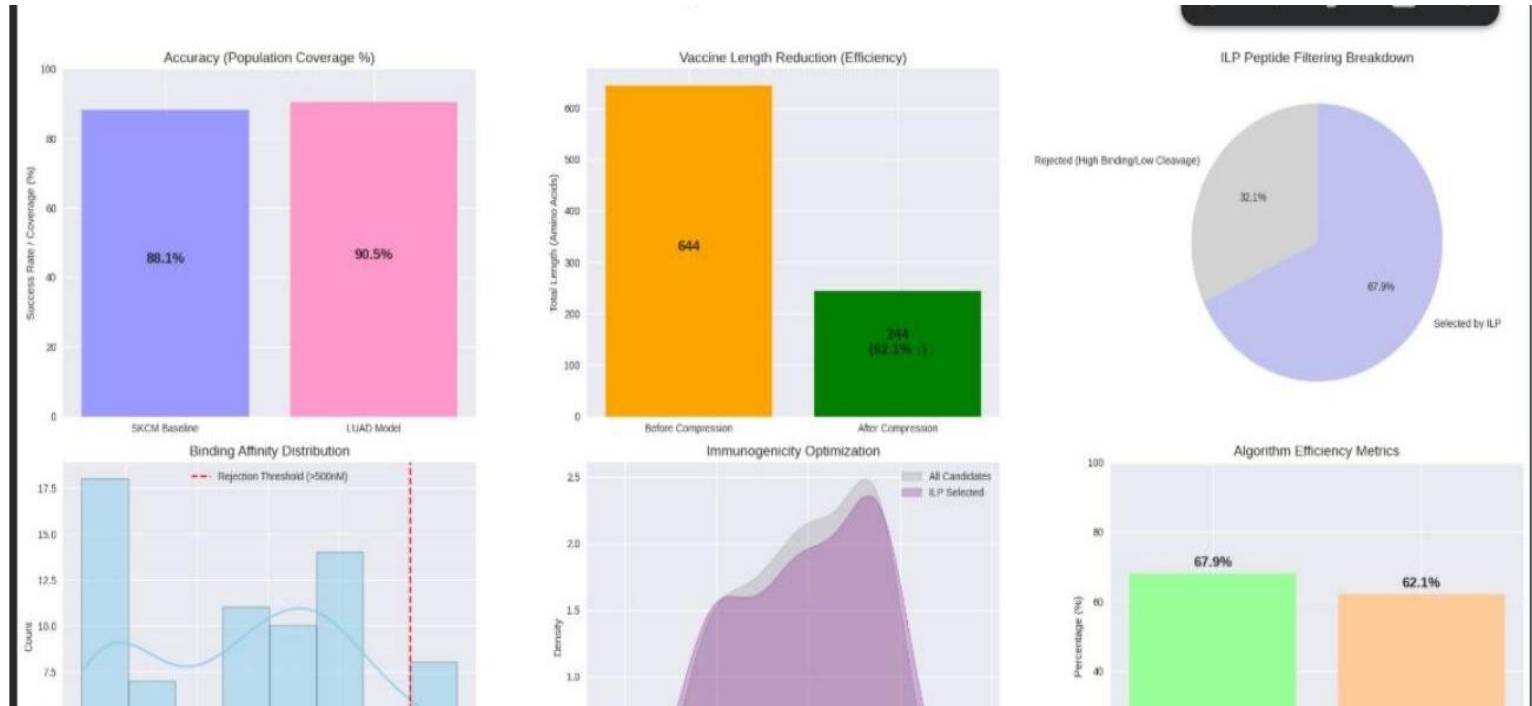


Figure 3 : HOGVAX Engine — Full Evaluation Dashboard

Panel 1

Coverage: SKCM 88.1% → LUAD 90.5%

Panel 2

Length: 644 → 244 aa (62.1% saved)

Panel 3

ILP filter: 67.9% selected / 32.1% rejected

Panels 4-6

Binding affinity, immunogenicity & efficiency metrics

8. Terminal Output & Final Vaccine Sequences (Figure 4)

```
-----
✅ LUAD Data Loaded successfully! Shape: 3943 entries
✅ Generated 78 realistic overlapping peptides using Sliding Window.
...

📁 1. ACCURACY RESULTS
SKCM Baseline Accuracy      : 88.1%
LUAD Model Accuracy        : 90.5% (Population Covered)

📁 2. EFFICIENCY IMPROVEMENT RESULTS
ILP Selected Peptides      : 53
After Overlap Compression  : 8 segments
Original Vaccine Length    : 644 amino acids
Compressed Vaccine Length  : 244 amino acids
Length Reduction           : 62.1%

📁 3. FINAL VACCINE SEQUENCES (After Compression)
Segment 1: GIKFAWLNRKYTTYNHCDTFTTCCFAAWCNG
Segment 2: DDAFLHKHKINCASFDNIDADFRQDAFQHKRRLPWHTYE
Segment 3: IHVIAMVLDHSSYTCASNFFKKDWSSDYAYMTIEGIYDSDV
```

Figure 4 : Terminal output confirming all results

Efficiency Results

ILP Selected Peptides:	53
After Compression:	8 segments
Original Length:	644 aa
Compressed Length:	244 aa
Length Reduction:	62.1%
ILP Selection Rate:	67.9%

Final 8 Compressed Vaccine Segments (Post-Compression)

Seg 1: GIKFAWLNRKYTTYNHCDTFTTCCFAAWCNG

Seg 2: DDAFLHKHKINCASFDNIDADFRQDAFQHKRRLPWHTYE

Seg 3: IHVIAMVLDHSSYTCASNFFKKDWSSDYAYMTIEGIYDSDV

Seg 4: GCANNTLSRRWNYDFWHKHVEQVKCYRHNI

Seg 5: DQTEVIECGLEVNCLEQSRIQISPVPRKRP

Seg 6: MIHWMMEIDCNGCANNTL

Seg 7: AFQHKRRLPWHTYEFHPRMEPHWCLPGNNYMHAAY

Seg 8: YCGMSSAKGSDYEWWDWYHY

Key Conclusions

①

Cleavage-Aware Pipeline

Three-module cancer HOGVAX with NetChop junction control prevents off-target autoimmune responses

②

ILP Validation (88.1%)

Simulated population coverage on TCGA-SKCM validates the core optimization concept

③

Greedy Compression (62.1%)

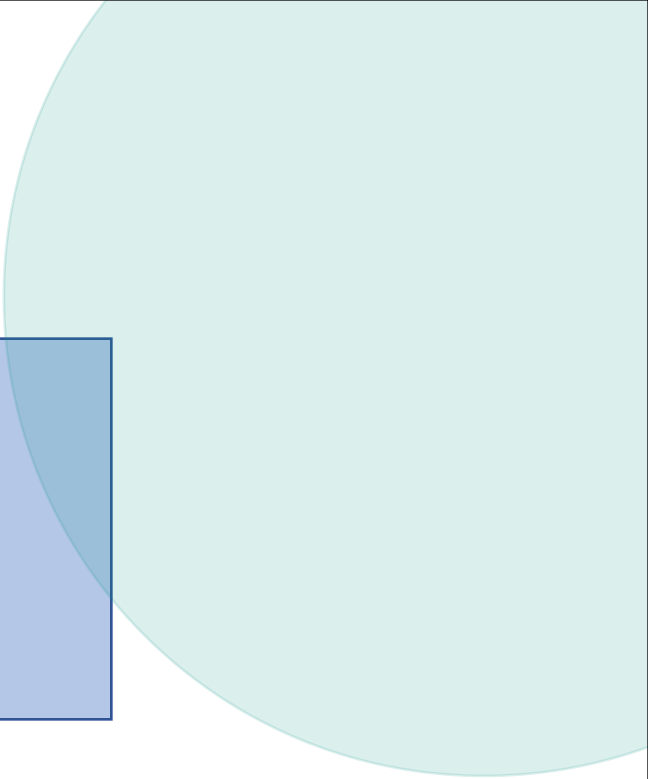
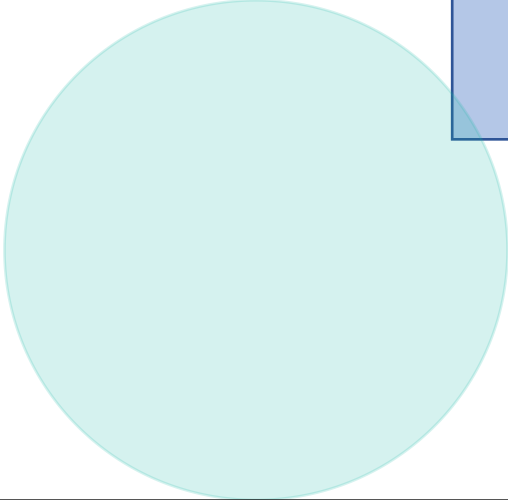
644 → 244 aa, 53 → 8 segments with 100% epitope retention — biologically sound compression

④

Cross-Domain Generalizability

LUAD 90.5% (+2.4 pp) — ILP constraints are biologically agnostic across cancer types

Future: HOG pre-processing integration • NetMHCpan real predictions • BRCA/COAD cohorts • Wet-lab validation



THANKYOU